TASK 2: Multiple Sequence Alignment using Clustal Omega
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Title -
Multiple Sequence Alignment and Conservation Analysis of Hemoglobin β - Chain Proteins Across Different Species

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Aim

To perform a multiple sequence alignment (MSA) of five homologous hemoglobin β -chain protein sequences from different species using Clustal Omega, and to identify and interpret conserved regions, functionally important residues, and sequence variations.

Protein Sequences Selected

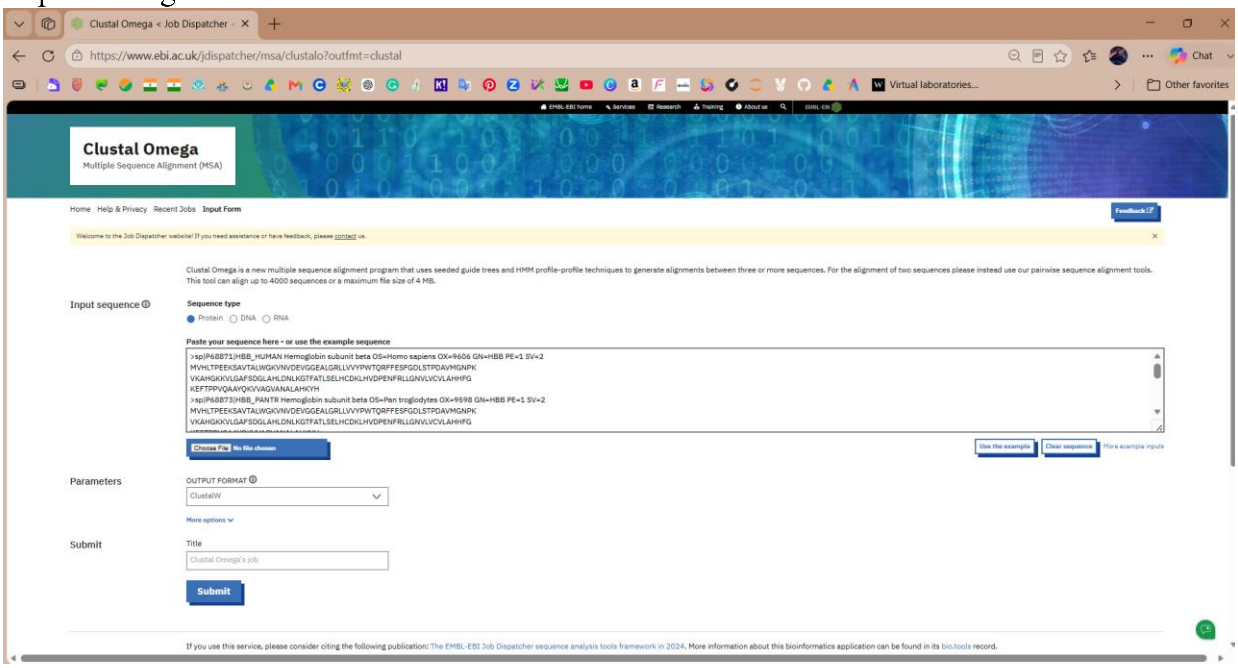
Five hemoglobin β-chain proteins were selected for the analysis.

No.	Organism	Protein	Accession	Length
1	*Homo sapiens*	Hemoglobin subunit beta	P68871	147 aa
2	*Pan troglodytes*	Hemoglobin subunit beta	P68873	147 aa
3	*Gorilla gorilla gorilla*	Hemoglobin subunit beta	XP_018891709.1	147 aa
4	*Mus musculus*	Hemoglobin subunit beta-1	P02088	147 aa
5	*Equus caballus*	Globin A1	F6RDD3	147 aa

All five sequences were protein sequences belonging to the hemoglobin/globin family and were suitable for comparative multiple sequence alignment.

Methodology

The protein sequences were obtained from UniProt and NCBI in FASTA format. The five FASTA sequences were combined and submitted to EMBL-EBI Clustal Omega for multiple sequence alignment.



The resulting alignment was examined for:
* Identical amino-acid residues
* Highly conserved regions
* Conservative amino-acid substitutions
* Variable regions
* Functionally important conserved residues

In the Clustal Omega alignment:

- * represents an identical amino acid in all sequences.
- : represents a strongly conserved/conservative substitution.
- . represents a weaker similarity between amino acids.

Results

Multiple Sequence Alignment

The five hemoglobin β -chain sequences were successfully aligned using Clustal Omega.

Highly conserved regions

```
sp|P02088|HBB1_MOUSE
MVHLTDAEKAAVSCLWGKVNSDEVGGEALGRLLVVYPWTQRYFDSFGDLSSASAIMGNAK
tr|F6RDD3|F6RDD3_HORSE
MVQLSGEEKAAVLALWDKVNEEEVGGEALGRLLVVYPWTQRFFDSFGDLSNPGAVMGNPK
sp|P68871|HBB_HUMAN
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
sp|P68873|HBB_PANTR
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
XP_018891709.1
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
**:*:  **:** .**.*** :*******************:*:******. .*:*** *

sp|P02088|HBB1_MOUSE
VKAHGKKVITAFNDGLNHLDSLKGTFASLSELHCDKLHVDPENFRLLGNMIVIVLGHHLG
tr|F6RDD3|F6RDD3_HORSE
VKAHGKKVLHSFGEGVHHLDNLKGTFAALSELHCDKLHVDPENFRLLGNVLVVVLARHFG
sp|P68871|HBB_HUMAN
VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG
sp|P68873|HBB_PANTR
VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG
XP_018891709.1
VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFKLLGNVLVCVLAHHFG
********: :*.:*: ***.******:****************:****::* **.:*:*

sp|P02088|HBB1_MOUSE        KDFTPAAQAAFQKVVAGVATALAHKYH
tr|F6RDD3|F6RDD3_HORSE      KDFTPELQASYQKVVAGVANALAHKYH
sp|P68871|HBB_HUMAN         KEFTPPVQAAYQKVVAGVANALAHKYH
sp|P68873|HBB_PANTR         KEFTPPVQAAYQKVVAGVANALAHKYH
XP_018891709.1              KEFTPPVQAAYQKVVAGVANALAHKYH
                            *:***  **::********.*******
```

Functionally important histidines

```
sp|P02088|HBB1_MOUSE
MVHLTDAEKAAVSCLWGKVNSDEVGGEALGRLLVVYPWTQRYFDSFGDLSSASAIMGNAK
tr|F6RDD3|F6RDD3_HORSE
MVQLSGEEKAAVLALWDKVNEEEVGGEALGRLLVVYPWTQRFFDSFGDLSNPGAVMGNPK
```

```
sp|P68871|HBB_HUMAN
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
sp|P68873|HBB_PANTR
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
XP_018891709.1
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
**:*:   **:** .**.*** :******************:*:******. .*:*** *

sp|P02088|HBB1_MOUSE
VKAHGKKVITAFNDGLNHLDSLKGTFASLSELHCDKLHVDPENFRLLGNMIVIVLGHHLG
tr|F6RDD3|F6RDD3_HORSE
VKAHGKKVLHSFGEGVHHLDNLKGTFAALSELHCDKLHVDPENFRLLGNVLVVVLARHFG
sp|P68871|HBB_HUMAN
VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG
sp|P68873|HBB_PANTR
VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG
XP_018891709.1
VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFKLLGNVLVCVLAHHFG
********: :*.:*: ***.******:****************:****::* **.:*:*

sp|P02088|HBB1_MOUSE        KDFTPAAQAAFQKVVAGVATALAHKYH
tr|F6RDD3|F6RDD3_HORSE      KDFTPELQASYQKVVAGVANALAHKYH
sp|P68871|HBB_HUMAN         KEFTPPVQAAYQKVVAGVANALAHKYH
sp|P68873|HBB_PANTR         KEFTPPVQAAYQKVVAGVANALAHKYH
XP_018891709.1              KEFTPPVQAAYQKVVAGVANALAHKYH
                            *:***  **::********.*******
```

Conservative substitution

```
sp|P02088|HBB1_MOUSE
MVHLTDAEKAAVSCLWGKVNSDEVGGEALGRLLVVYPWTQRYFDSFGDLSSASAIMGNAK
                                          ↑
                                          Y(Conservative F→Y substitution)
tr|F6RDD3|F6RDD3_HORSE
MVQLSGEEKAAVLALWDKVNEEEVGGEALGRLLVVYPWTQRFFDSFGDLSNPGAVMGNPK
sp|P68871|HBB_HUMAN
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
sp|P68873|HBB_PANTR
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
XP_018891709.1
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
**:*:   **:** .**.*** :******************:*:******. .*:*** *

sp|P02088|HBB1_MOUSE
VKAHGKKVITAFNDGLNHLDSLKGTFASLSELHCDKLHVDPENFRLLGNMIVIVLGHHLG
tr|F6RDD3|F6RDD3_HORSE
VKAHGKKVLHSFGEGVHHLDNLKGTFAALSELHCDKLHVDPENFRLLGNVLVVVLARHFG
sp|P68871|HBB_HUMAN
VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG
sp|P68873|HBB_PANTR
VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG
XP_018891709.1
VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFKLLGNVLVCVLAHHFG
********: :*.:*: ***.******:****************:****::* **.:*:*

sp|P02088|HBB1_MOUSE        KDFTPAAQAAFQKVVAGVATALAHKYH
tr|F6RDD3|F6RDD3_HORSE      KDFTPELQASYQKVVAGVANALAHKYH
sp|P68871|HBB_HUMAN         KEFTPPVQAAYQKVVAGVANALAHKYH
```

```
sp|P68873|HBB_PANTR          KEFTPPVQAAYQKVVAGVANALAHKYH
XP_018891709.1               KEFTPPVQAAYQKVVAGVANALAHKYH
                             *:***   **::********.*******
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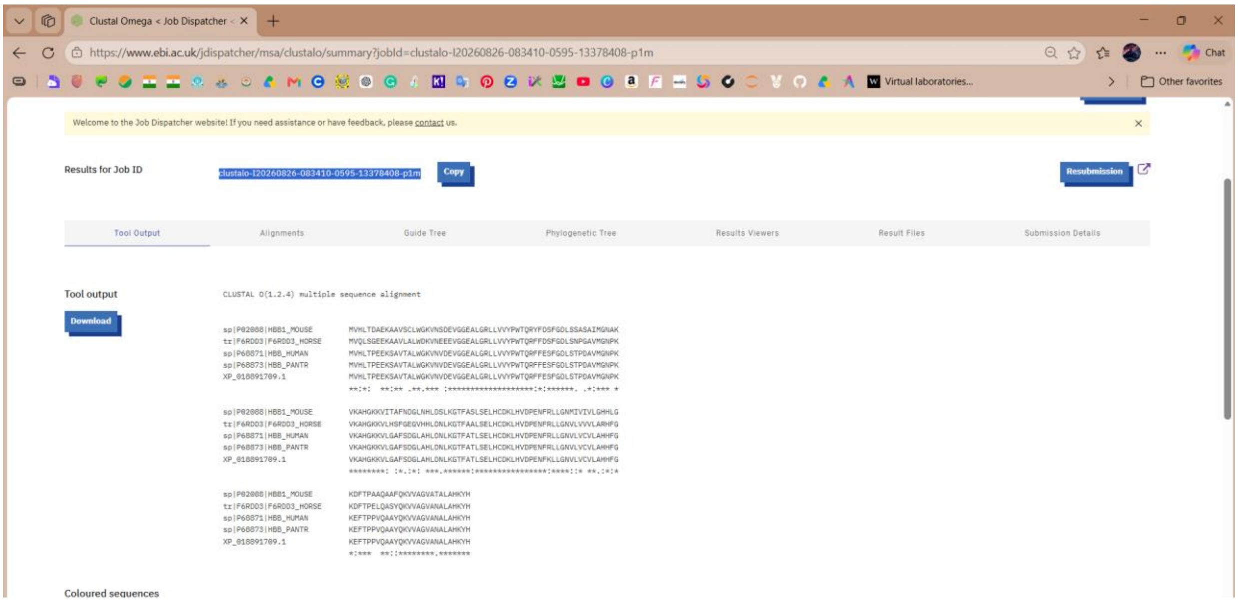

Multiple sequence alignment of five hemoglobin β-chain protein sequences generated using Clustal Omega. Yellow indicates highly conserved regions, green indicates functionally important histidine residues, and blue indicates the conservative F→Y substitution observed in the mouse sequence.

Conserved Regions

The alignment revealed several highly conserved regions among the five sequences.
One prominent region is:
GGEALGRLLVVYPWTQRFF
This region is identical in human, chimpanzee, gorilla and horse sequences. The mouse sequence contains a conservative substitution:
Human: GGEALGRLLVVYPWTQRFF
Mouse: GGEALGRLLVVYPWTQRYF
Thus, the region can be described as a highly conserved region with a conservative F→Y substitution in mouse.
Another highly conserved region identified in the alignment is:
SELHCDKLHVDPENFRLLGNV
A conserved C-terminal region was also observed:
QKVVAGVANALAHKYH
The high degree of conservation suggests that these regions are important for maintaining the structural and functional properties of hemoglobin β-chain proteins.

Conserved Histidine Residues

Histidine-containing regions were specifically examined because conserved histidine residues are associated with the heme-binding environment of hemoglobin.
The alignment contains the region:
SELHCDKLH
and another histidine-rich region:
GNVLVCVLAHHFG

These conserved histidine residues are relevant to the structural environment surrounding the heme group.

Conservative Substitution in Mouse

An interesting sequence variation was observed between the human and mouse ?-globin sequences:
Human: ...VVYPWTQRFF...
Mouse: ...VVYPWTQRYF...
Here, one phenylalanine (F) in human is replaced by tyrosine (Y) in mouse.
This is considered a conservative substitution because phenylalanine and tyrosine are both aromatic amino acids with similar chemical characteristics.
This observation demonstrates that sequence variation can occur while retaining chemically similar amino-acid properties.

Interpretation

The multiple sequence alignment demonstrates a high degree of conservation among the five hemoglobin β-chain sequences.
The human, chimpanzee, and gorilla sequences show particularly high similarity. This is consistent with the close evolutionary relationship among these primates. The mouse and horse sequences contain a greater number of amino-acid substitutions, but several regions remain strongly conserved.
The presence of highly conserved regions suggests that these parts of the protein are under evolutionary constraint. Changes in structurally or functionally important regions may affect protein stability or function, which can explain why such regions remain conserved across species.
The F→Y substitution observed in the mouse sequence provides an example of a conservative amino-acid change. Although the residue differs, the replacement involves chemically similar amino acids.
The conserved histidine-containing regions are also significant because histidine residues contribute to the heme-binding environment of hemoglobin. Therefore, conservation of these residues is consistent with their functional importance.
Overall, the MSA demonstrates that hemoglobin β-chain proteins maintain substantial sequence conservation while allowing specific evolutionary substitutions.

Phylogenetic Analysis

A sequence-based tree was generated from the aligned sequences to visualize their relationships.

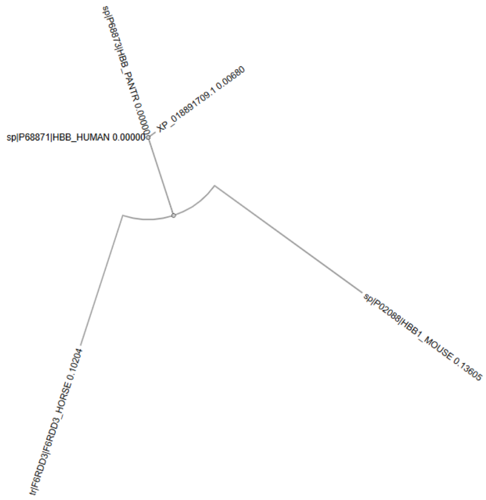

Figure 3. Sequence-based phylogenetic tree generated from the multiple sequence alignment of the five hemoglobin β-chain proteins.

The tree shows that the human and chimpanzee sequences are most closely grouped, followed by the gorilla sequence. The mouse and horse sequences are positioned farther from the human/chimpanzee/gorilla group, consistent with the greater sequence differences observed in the MSA.

The tree represents relationships based on the selected protein sequences and should therefore be interpreted as a sequence-based relationship rather than a complete evolutionary history of the species.

Conclusion

Multiple sequence alignment of five hemoglobin β-chain proteins was successfully performed using Clustal Omega. The analysis revealed extensive conservation across the selected species, particularly among the human, chimpanzee, and gorilla sequences.

Several highly conserved regions and functionally important histidine residues were identified.

A conservative F→Y substitution was also observed in the mouse sequence, demonstrating evolutionary variation while retaining similar amino-acid properties.

The results show that MSA is a useful computational approach for identifying conserved regions and sequence variations that can provide insight into the structural and functional importance of proteins.